



Soroush Haghi

Computer Engineering | Software & AI | Quantum Technologies

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PROFESSIONAL PROFILE

M.Sc. student in Quantum Technologies at TU Braunschweig with a B.Sc. in Computer Engineering (AI specialization). Practical work is strongest in Python-based machine learning and computer vision, sensor and signal processing, MATLAB, and technical software/automation systems. Academic depth spans quantum information and computing, quantum communication and photonics, semiconductor/device technologies, and information/coding theory; project experience also includes embedded systems and industry-university engineering.

EDUCATION

M.Sc. Quantum Technologies in Electrical and Computer Engineering

Technische Universität Braunschweig, Germany | 66/120 ECTS completed

Oct 2024 - Present

Next: Internship & Thesis

Academic focus: Quantum Information & Computing • Quantum Communication & Photonics • Semiconductor & Quantum Devices • Information & Coding Theory

Selected academic work: QKD satellite communication • Nonlinear photonics analytical coursework • Coding-theory computational work • Semiconductor crystal defects

B.Sc. Computer Engineering (AI specialization)

2018 - Feb 2024

Azad University, Mashhad Branch, Iran

Academic focus: Software, Programming & Algorithms • AI, Machine Learning & Computer Vision • Embedded Systems & Hardware • Databases & Information Systems

EXPERIENCE

Iran Khodro (IKCO)

Apr 2023 - Sep 2023

Pouyesh Scheme - University Project Member

- Participated through the university in a MATLAB computer-vision vehicle-detection project.
- Project participation led to an invitation to work from Iran Khodro.

Ferdowsi University of Mashhad, Iran

Mar 2023 - Aug 2023

Project Contributor - Anodyne (University-Based Startup)

- Supported project coordination, external presentations, stakeholder communication, and outreach for a dental imaging initiative.
- Assisted with attendee coordination, on-site project presentation, promotional activities, and professional networking during dental and medical industry events.

Azad University, Mashhad Branch, Iran

Teaching Assistant - Theory of Formal Languages and Automata

Sep 2023 - Jan 2024

- Led weekly exercise and problem-solving sessions for approximately 20 students.
- Corrected and evaluated assignments and exams within the supervised course.

Volunteer IT Support

Oct 2022 - Aug 2023

- Continued supporting the university's IT team voluntarily alongside my studies after completing the internship.
- Progressed from central IT support to on-site troubleshooting and repair across university departments.

IT Support Intern

Jul 2022 - Sep 2022

- Diagnosed and repaired desktop systems, including component replacement/upgrades and Windows installation/configuration.
- Supported physical network infrastructure, including rack/cabling work, plus local-network and Wi-Fi setup/troubleshooting.

TECHNICAL SKILLS

Programming

Python; C; C++; MATLAB

Machine Learning & Computer Vision

PyTorch; scikit-learn

Scientific Computing & Signal Processing

NumPy; SciPy; pandas; Matplotlib

Embedded Systems

ATmega32

Development Tools & Databases

Git; GitHub; GitLab; SQL

SELECTED TECHNICAL PROJECTS

AI SYSTEMS & WORKFLOW AUTOMATION

Career OS - Privacy-First AI Context & Knowledge System ([Link](#))

Google Apps Script; Google Drive API; Gemini; Groq; GitHub

- Designed and co-developed a privacy-first system for shared AI context, evidence tracking, source ingestion, provenance-aware synthesis, queue/retry processing, and CI privacy checks.

MACHINE LEARNING & COMPUTER VISION

MRI Segmentation Inference Dashboard ([Link](#))

PyTorch; Streamlit

- Integrated pretrained segmentation inference with preprocessing, weight loading, masks, overlays, and area visualization.

Gas Classification Pipeline and Simulation Dashboard ([Link](#))

Python; scikit-learn; RandomForest; Streamlit

- Built statistical sensor-window features and a six-gas classification dashboard with class-probability output.

Activity Recognition ML Comparison ([Link](#))

Python; PyTorch; scikit-learn; Poetry

- Compared RandomForest and 1D-CNN approaches for sensor activity recognition and configured lint/test quality checks.

Day/Night Vehicle Detection ([Link](#))

MATLAB; Computer Vision; Image Processing

- Developed a road-intersection pipeline with day/night routing, foreground detection, morphology, and region-based vehicle filtering.

SIGNAL & IMAGE PROCESSING

Hybrid Image Generation ([Link](#)) | Stereo Sound Direction ([Link](#))

MATLAB; Image and Signal Processing

- Combined low/high spatial-frequency image content; separately analyzed stereo-audio window energy for dominant left/right direction.

EMBEDDED SYSTEMS

ATmega32 Digital Clock ([Link](#)) | ATmega32 Digital Safe ([Link](#))

Embedded C; ATmega32; 7-segment display; matrix keypad; LCD

- Implemented coursework projects for a multiplexed digital clock and a keypad/password digital safe.

SELECTED ACADEMIC TECHNICAL WORK

Coding & Information Theory

MATLAB/group and graded work on cyclic and block codes, Hamming distance/correction radius, bit-error behavior versus SNR, prefix codes, relative entropy, Markov processes, and typical sets.

Nonlinear Photonics

Analytical work on dispersion, phase matching, harmonic generation, nonlinear optical effects, fiber nonlinearities, and gain/threshold calculations.

HONORS

Iran Khodro (IKCO) Pouyesh: Ranked 1st of 50 participants	(Top 2%)
B.Sc. Computer Engineering: Ranked 5th of 131 students	(Top 4%)
Iran National University Entrance Exam (Konkoor) 2018: ~10,000 of ~140,000	(Top 7%)
Khorasan Razavi selective-school Entrance Exam 2014: 12 of ~27,000	(Top 1%)

LANGUAGES & CERTIFICATES

Languages	German - A2 course completed; exam result pending; next: B1.1 English - TOEFL iBT 95/120 Persian - Native
Training / Certificates	Sharif University of Technology - Quantum Programming Language (in progress) 2026 Ariaquanta Institute - Quantum Machine Learning 2025 Kaggle - Machine Learning; Introduction to Programming Using Python 2025 Azad University, Mashhad - Data & ML Bootcamp 96 hours 2023 Jahad Daneshgahi, Khorasan Razavi - Python Programming 40 hours 2021